

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

QUIZ-1

DURATION: 35 MINUTES

SUMMER SEMESTER, 2022-2023

FULL MARKS: 15

**Math 4641: Numerical Methods**

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Suppose,  $R$  and  $k$  are arbitrary numbers such that  $R > 0$  and  $k \neq 0$ . Prove that the Newton-Raphson formula for finding the  $k$ th root of  $R$ , which is mathematically denoted as  $\sqrt[k]{R}$ , is as shown in Equation-1. 3  
(CO2)  
(PO1)

$$x_{i+1} = \frac{1}{k} \left[ (k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad (1)$$

2. a) What are significant digits? What is the relationship between the number of significant digits that are at least correct ( $m$ ) in an approximation and the relative approximate error ( $\epsilon_a$ )? 1  
(CO2)  
(PO1)
- b) Determine the *minimum* number of significant digits in each of these numerical values — 3  
(CO4)  
(PO1)
- (a) 69.0                                      (b) 69,420                                      (c)  $6.94200 \times 10^{911}$
- (d) 69.00420                                      (e) 069420                                      (f) 0.069420

3. “If a function  $f(x)$  does **not** change its sign between two points  $x_l$  and  $x_u$ , then there may or may not be any root of the equation  $f(x) = 0$  between them.” — Do you agree? Justify your answer. 3  
(CO3)  
(PO2)

4. The *Plastic Ratio*, often denoted by the Greek letter  $\rho$  (rho), is the ratio involving three quantities  $a$ ,  $b$ , and  $c$ , such that  $a > b > c > 0$ , if they satisfy the equality  $\frac{a}{b} = \frac{b+c}{a} = \frac{b}{c}$ . The constant  $\rho$  is equal to the unique real root of the cubic equation  $x^3 = x + 1$ . The numerical value of that root is, 5  
(CO2)  
(PO1)

$$x_{root} = \rho = \sqrt[3]{\frac{9 + \sqrt{69}}{18}} + \sqrt[3]{\frac{9 - \sqrt{69}}{18}} = 1.324717957 \dots$$

Apply the *Regula Falsi* method, also known as the False Position method, to estimate the value of the plastic ratio  $\rho$ . The initial upper and lower guesses are  $x_l = -1$  and  $x_u = 2$  respectively. Conduct 4 iterations, and find the relative approximate error ( $|\epsilon_a|\%$ ) and the number of significant digits that are at least correct ( $m$ ), at the end of each iteration. Fill out the following table after performing the necessary calculations.

| Iteration | $x_l$ | $x_u$ | $x_r$ | $ \epsilon_a \%$ | $m$ | $f(x_r)$ |
|-----------|-------|-------|-------|------------------|-----|----------|
| 1         | -1    | 2     |       | —                | —   |          |
| 2         |       |       |       |                  |     |          |
| 3         |       |       |       |                  |     |          |
| 4         |       |       |       |                  |     |          |